TABLE 8.3 What Is Your Retirement's Probability of Sustainability?

α/β	0.25	0.50	0.75	1.00	1.25	1.50	1.75	2.00	2.25	2.50	2.75
4.00	100%	100%	99%	98%	96%	93%	90%	86%	81%	76%	70%
3.75	100%	100%	99%	97%	95%	91%	87%	82%	77%	71%	65%
3.50	100%	99%	98%	96%	93%	89%	84%	78%	72%	66%	60%
3.25	100%	99%	97%	94%	90%	85%	79%	73%	67%	60%	54%
3.00	100%	99%	96%	92%	87%	81%	74%	68%	61%	54%	48%
2.75	100%	98%	94%	89%	83%	76%	69%	62%	55%	48%	42%
2.50	99%	96%	91%	85%	78%	70%	62%	55%	48%	42%	36%
2.25	99%	94%	88%	80%	72%	63%	55%	48%	41%	35%	30%
2.00	97%	91%	83%	74%	64%	56%	48%	41%	34%	29%	24%
1.75	95%	86%	76%	66%	56%	48%	40%	33%	28%	23%	19%
1.50	92%	80%	68%	57%	48%	39%	32%	26%	21%	17%	14%
1.25	86%	72%	59%	47%	38%	31%	24%	19%	15%	12%	10%
1.00	78%	61%	47%	37%	29%	22%	17%	14%	11%	8%	6%
0.75	65%	47%	35%	26%	20%	15%	11%	9%	6%	5%	4%
0.50	48%	32%	22%	16%	11%	8%	6%	5%	3%	3%	2%

Source: Moshe Milevsky and the IFID Centre, 2008, based on methodology from Milevsky, M., and Robinson, C., "A Sustainable Spending Rate Without Simulation." *Financial Analysts Journal*. Nov/Dec 2005, 61(6).

Stay with me here. According to the mathematical equations I listed earlier, my retirement alpha is equal to 3 units and my retirement beta is equal to 1.74 units. These are the intermediary ingredients for the main formula.

Finally, I take these two values and look up the retirement sustainability probability from Table 8.3. In the alpha = 3 and beta = 1.74 case, the retirement sustainability probability is approximately 74%. Thus, if I take early retirement today at the age of 40, my odds don't look very good. I will be spending too much, living too long, or simply not earning enough to achieve my sustainability goals.

In contrast, if I were to take early retirement and only spend 2% per year adjusted for inflation, which is $20,000 per $1,000,000 nest egg, the situation would look better. In this case my retirement alpha would be the same 3 units, but my retirement beta would be reduced to 0.70 units. My success ratio would increase to more than 96%. To